DOCTYPE Definition and Usage

The <!DOCTYPE> declaration must be the very first thing in your HTML document, before the <html> tag.

The <!DOCTYPE> declaration is not an HTML tag; it is an instruction to the web browser about what version of HTML the page is written in.

In HTML 4.01, the <!DOCTYPE> declaration refers to a DTD, because HTML 4.01 was based on SGML. The DTD specifies the rules for the markup language, so that the browsers render the content correctly.

HTML5 is not based on SGML, and therefore does not require a reference to a DTD.

Tip: Always add the <!DOCTYPE> declaration to your HTML documents, so that the browser knows what type of document to expect.

Common DOCTYPE Declarations

HTML 5

```
<!DOCTYPE html>
```

HTML 4.01 Strict

This DTD contains all HTML elements and attributes, but does NOT INCLUDE presentational or deprecated elements (like font). Framesets are not allowed.

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"
"http://www.w3.org/TR/html4/strict.dtd">
```

HTML 4.01 Transitional

This DTD contains all HTML elements and attributes, INCLUDING presentational and deprecated elements (like font). Framesets are not allowed.

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
```

HTML 4.01 Frameset

This DTD is equal to HTML 4.01 Transitional, but allows the use of frameset content.

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Frameset//EN"
"http://www.w3.org/TR/html4/frameset.dtd">
```

Meta Tags

```
<head>
<meta charset="UTF-8">
<meta name="description" content="Free Web tutorials">
<meta name="keywords" content ="HTML, CSS, XML, Java Script"> //20 max
<meta name ="author" Content="Chandan Mukherjee">
<meta http-equiv ="content-type" content="text/html; charset=UTF-8">
</head>
```

The Unicode Consortium

The Unicode Consortium develops the Unicode Standard. Their goal is to replace the existing character-sets with its standard Unicode Transformation Format (UTF).

The Unicode Standard has become a success and is implemented in XML, Java, ECMAScript (JavaScript), LDAP, CORBA 3.0, WML, etc. The Unicode standard is also supported in many operating systems and all modern browsers.

The Unicode Consortium cooperates with the leading standards development organizations, like ISO, W3C, and ECMA.

Unicode can be implemented by different character-sets. The most commonly used encodings are UTF-8 and UTF-16:

Character-set	Description
UTF-8	A character in UTF8 can be from 1 to 4 bytes long. UTF-8 can represent any character in the Unicode standard. UTF-8 is backwards compatible with ASCII. UTF-8 is the preferred encoding for e-mail and web pages
UTF-16	16-bit Unicode Transformation Format is a variable-length character encoding for Unicode, capable of encoding the entire Unicode repertoire. UTF-16 is used in major operating systems and environments, like Microsoft Windows 2000/XP/2003/Vista/CE and the Java and .NET byte code environments

Tip: The first 256 characters of Unicode character-sets correspond to the 256 characters of ISO-8859-1.

Tip: All HTML 4 processors already support UTF-8, and all XHTML and XML processors support UTF-8 and UTF-16!

HTML5 is the next major revision of the HTML standard superseding HTML 4.01, XHTML 1.0, and XHTML 1.1. HTML5 is a

standard for structuring and presenting content on the World Wide Web.

HTML5 is a cooperation between the World Wide Web Consortium (W3C) and the Web Hypertext Application Technology Working Group (WHATWG).

The new standard incorporates features like video playback and drag-and-drop that have been previously dependent on third-party browser plug-ins such as Adobe Flash, Microsoft Silverlight, and Google Gears.

Browser Support:

The latest versions of Apple Safari, Google Chrome, Mozilla Firefox, and Opera all support many HTML5 features and Internet Explorer 9.0 will also have support for some HTML5 functionality.

The mobile web browsers that come pre-installed on iPhones, iPads, and Android phones all have excellent support for HTML5.

New Features:

HTML5 introduces a number of new elements and attributes that helps in building a modern websites. Following are great features introduced in HTML5.

- **New Semantic Elements:** These are like <header>, <footer>, and <section>.
- **Forms 2.0:** Improvements to HTML web forms where new attributes have been introduced for <input> tag.
- **Persistent Local Storage:** To achieve without resorting to third-party plugins.
- **WebSocket:** A next-generation bidirectional communication technology for web applications.
- **Server-Sent Events:** HTML5 introduces events which flow from web server to the web browsers and they are called Server-Sent Events (SSE).
- **Canvas:** This supports a two-dimensional drawing surface that you can program with JavaScript.
- **Audio & Video:** You can embed audio or video on your web pages without resorting to third-party plugins.
- **Geolocation:** Now visitors can choose to share their physical location with your web application.
- **Microdata:** This lets you create your own vocabularies beyond HTML5 and extend your web pages with custom semantics.
- **Drag and drop:** Drag and drop the items from one location to another location on the same webpage.

HTML 5 does not have the same syntax rules as XHTML where we needed lower case tag names, quoting our attributes,an attribute had to have a value and to close all empty elements.

But HTML5 is coming with lots of flexibility and would support the followings:

- Uppercase tag names.
- Quotes are optional for attributes.
- Attribute values are optional.
- Closing empty elements are optional.

The DOCTYPE:

DOCTYPEs in older versions of HTML were longer because the HTML language was SGML based and therefore required a reference to a DTD.

HTML 5 authors would use simple syntax to specify DOCTYPE as follows:

```
<!DOCTYPE html>
```

All the above syntax is case-insensitive.

Character Encoding:

HTML 5 authors can use simple syntax to specify Character Encoding as follows:

```
<meta charset="UTF-8">
```

All the above syntax is case-insensitive.

The <script> tag:

It's common practice to add a type attribute with a value of "text/javascript" to script elements as follows:

```
<script type="text/javascript" src="scriptfile.js"></script>
```

HTML 5 removes extra information required and you can use simply following syntax:

```
<script src="scriptfile.js"></script>
```

The <link> tag:

So far you were writing <link> as follows:

```
<link rel="stylesheet" type="text/css" href="stylefile.css">
```

HTML 5 removes extra information required and you can use simply following syntax:

```
<link rel="stylesheet" href="stylefile.css">
```

HTML5 tag names are **case insensitive** and may be written in all uppercase or mixed case, although the most common convention is to stick with lowercase.

Most of the elements contain some content like ... contains a paragraph. Some elements, however, are forbidden from containing any content at all and these are known as void elements. For example, br, hr, link and meta etc.

Deprecated Tags:

The following elements are not available in HTML5 anymore and their function is better handled by CSS:

Tags (Elements)	Description
<acronym>	Defines an acronym
<applet>	Defines an applet
<basefont>	Defines an base font for the page.
<big>	Defines big text
	Defines centered text
<dir>	Defines a directory list
<font>	Defines text font, size, and color
<frame>	Defines a frame
<frameset>	Defines a set of frames
<isindex>	Defines a single-line input field
<noframes>	Defines a noframe section
<s>	Defines strikethrough text
<strike>	Defines strikethrough text
<tt>	Defines teletype text
<u>	Defines underlined text

HTML5 New Tags (Elements)

The following tags (elements) have been introduced in HTML5:

Tags (Elements)	Description
<article>	Represents an independent piece of content of a document, such as a blog entry or newspaper article
<aside >	Represents a piece of content that is only slightly related to the rest of the page.
<audio>	Defines an audio file.
<canvas>	This is used for rendering dynamic bitmap graphics on the fly, such as graphs or games.
<command>	Represents a command the user can invoke.
<datalist>	Together with the a new list attribute for input can be used to make combo boxes
<details>	Represents additional information or controls which the user can obtain on demand
<embed>	Defines external interactive content or plugin.
<figure>	Represents a piece of self-contained flow content, typically referenced as a single

	unit from the main flow of the document.
<footer>	Represents a footer for a section and can contain information about the author, copyright information, et cetera.
<header>	Represents a group of introductory or navigational aids.
<hgroup>	Represents the header of a section.
<keygen>	Represents control for key pair generation.
<mark>	Represents a run of text in one document marked or highlighted for reference purposes, due to its relevance in another context.
<meter>	Represents a measurement, such as disk usage.
<nav>	Represents a section of the document intended for navigation.
<output>	Represents some type of output, such as from a calculation done through scripting.
<progress>	Represents a completion of a task, such as downloading or when performing a series of expensive operations.
<ruby>	Together with <rt> and <rp> allow for marking up ruby annotations.
<section>	Represents a generic document or application section
<time>	Represents a date and/or time.
<video>	Defines a video file.
<wbr>	Represents a line break opportunity.

New types for <input> tag for HTML 5:

The input element's type attribute now has the following new values:

Type	Description
color	Color selector, which could be represented by a wheel or swatch picker
date	Selector for calendar date
datetime-local	Date and time display, with no setting or indication for time zones
datetime	Full date and time display, including a time zone.
email	Input type should be an email.
month	Selector for a month within a given year
number	A field containing a numeric value only
range	Numeric selector within a range of values, typically visualized as a slider
search	Term to supply to a search engine. For example, the search bar atop a browser.

tel	Input type should be telephone number.
time	Time indicator and selector, with no time zone information
url	Input type should be URL type.
week	Selector for a week within a given year

The <input> element in HTML4

HTML4 input elements use the **type** attribute to specify the data type.HTML4 provides following types:

Type	Description
text	A free-form text field, nominally free of line breaks.
password	A free-form text field for sensitive information, nominally free of line breaks.
checkbox	A set of zero or more values from a predefined list.
radio	An enumerated value.
submit	A free form of button initiates form submission.
file	An arbitrary file with a MIME type and optionally a file name.
image	A coordinate, relative to a particular image's size, with the extra semantic that it must be the last value selected and initiates form submission.
hidden	An arbitrary string that is not normally displayed to the user.
select	An enumerated value, much like the radio type.
textarea	A free-form text field, nominally with no line break restrictions.
button	A free form of button which can initiates any event related to button.

HTML5 introduces two mechanisms, similar to HTTP session cookies, for storing structured data on the client side and to overcome following drawbacks.

- Cookies are included with every HTTP request, thereby slowing down your web application by transmitting the same data.
- Cookies are included with every HTTP request, thereby sending data unencrypted over the internet.
- Cookies are limited to about 4 KB of data . Not enough to store required data.

The two storages are **session storage** and **local storage** and they would be used to handle different situations.

The latest versions of pretty much every browser supports HTML5 Storage including Internet Explorer.

Session Storage:

The *Session Storage* is designed for scenarios where the user is carrying out a single transaction, but could be carrying out multiple transactions in different windows at the same time.

Example:

For example, if a user buying plane tickets in two different windows, using the same site. If the site used cookies to keep track of which ticket the user was buying, then as the user clicked from page to page in both windows, the ticket currently being purchased would "leak" from one window to the other, potentially causing the user to buy two tickets for the same flight without really noticing.

HTML5 introduces the *sessionStorage* attribute which would be used by the sites to add data to the session storage, and it will be accessible to any page from the same site opened in that window ie session and as soon as you close the window, session would be lost.

Following is the code which would set a session variable and access that variable:

```
<!DOCTYPE HTML>
<html>
<body>

  <script type="text/javascript">
   if( sessionStorage.hits )
  {
      sessionStorage.hits = Number(sessionStorage.hits) +1;
   }
   else
  {
      sessionStorage.hits = 1;
   }
   document.write("Total Hits :" + sessionStorage.hits );
  </script>
  <p>Refresh the page to increase number of hits.</p>
  <p>Close the window and open it again and check the result.</p>

</body>
</html>
```

Local Storage:

The *Local Storage* is designed for storage that spans multiple windows, and lasts beyond the current session. In particular, Web applications may wish to store megabytes of user data, such as entire user-authored documents or a user's mailbox, on the client side for performance reasons.

Again, cookies do not handle this case well, because they are transmitted with every request.

Example:

HTML5 introduces the *localStorage* attribute which would be used to access a page's local storage area without no time limit and this local storage will be available whenever you would use that page.

Following is the code which would set a local storage variable and access that variable every time this page is accessed, even next time when you open the window:

```
<!DOCTYPE HTML>
<html>
```

```
<body>

  <script type="text/javascript">
    if( localStorage.hits ){
       localStorage.hits = Number(localStorage.hits) +1;
    }else{
       localStorage.hits = 1;
    }
    document.write("Total Hits :" + localStorage.hits );
  </script>
  <p>Refresh the page to increase number of hits.</p>
  <p>Close the window and open it again and check the result.</p>

</body>
</html>
```

Delete Web Storage:

Storing sensitive data on local machine could be dangerous and could leave a security hole.

The *Session Storage Data* would be deleted by the browsers immediately after the session gets terminated.

To clear a local storage setting you would need to call **localStorage.remove('key')**; where 'key' is the key of the value you want to remove. If you want to clear all settings, you need to call **localStorage.clear()** method.

```
<!DOCTYPE HTML>
<html>
<body>

  <script type="text/javascript">
    localStorage.clear();

    // Reset number of hits.
    if( localStorage.hits ){
       localStorage.hits = Number(localStorage.hits) +1;
    }else{
       localStorage.hits = 1;
    }
    document.write("Total Hits :" + localStorage.hits );
  </script>
  <p>Refreshing the page would not to increase hit counter.</p>
  <p>Close the window and open it again and check the result.</p>

</body>
</html>
```

HTML5 - Audio & Video

HTML5 features, include native audio and video support without the need for Flash.

The HTML5 <audio> and <video> tags make it simple to add media to a website. You need to set**src** attribute to identify the media source and include a controls attribute so the user can play and pause the media.

Embedding Video:

Here is the simplest form of embedding a video file in your webpage:

```
<video src="foo.mp4"  width="300" height="200" controls>
    Your browser does not support the <video> element.
</video>
```

The current HTML5 draft specification does not specify which video formats browsers should support in the video tag. But most commonly used video formats are:

1. **Ogg:** Ogg files with Thedora video codec and Vorbis audio codec.
2. **mpeg4:** MPEG4 files with H.264 video codec and AAC audio codec.

You can use <source> tag to specify media along with media type and many other attributes. A video element allows multiple source elements and browser will use the first recognized format:

Video Attribute Specification:

The HTML5 video tag can have a number of attributes to control the look and feel and various functionalities of the control:

Attribute	Description
autoplay	This boolean attribute if specified, the video will automatically begin to play back as soon as it can do so without stopping to finish loading the data.
autobuffer	This boolean attribute if specified, the video will automatically begin buffering even if it's not set to automatically play.
controls	If this attribute is present, it will allow the user to control video playback, including volume, seeking, and pause/resume playback.
height	This attribut specifies the height of the video's display area, in CSS pixels.
loop	This boolean attribute if specified, will allow video automatically seek back to the start after reaching at the end.
preload	This attribute specifies that the video will be loaded at page load, and ready to run. Ignored if autoplay is present.
poster	This is a URL of an image to show until the user plays or seeks.
src	The URL of the video to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed
width	This attribut specifies the width of the video's display area, in CSS pixels.

Embedding Audio:

HTML5 supports <audio> tag which is used to embed sound content in an HTML or XHTML document as follows.

```
<audio src="foo.wav" controls autoplay>
    Your browser does not support the <audio> element.
</audio>
```

The current HTML5 draft specification does not specify which audio formats browsers should support in the audio tag. But most commonly used audio formats are **ogg, mp3** and **wav**.

Geolocation getCurrentPosition() API

Description:

The getCurrentPosition method retrieves the current geographic location of the device. The location is expressed as a set of geographic coordinates together with information about heading and speed. The location information is returned in a Position object.

Syntax:

Here is the syntax of this method:

```
getCurrentPosition(showLocation, ErrorHandler, options);
```

Parameters:

Here is the detail of parameters:

- **showLocation** : This specifies the callback method that retrieves the location information. This method is called asynchronously with an object corresponding to the**Position** object which stores the returned location information.
- **ErrorHandler** : This optional paramter specifies the callback method that is invoked when an error occurs in processing the asynchronous call. This method is called with the**PositionError** object that stores the returned error information.
- **options** : This optional paramter specifies a set of options for retrieving the location information. You can specify (a) Accuracy of the returned location information (b) Timeout for retrieving the location information and (c) Use of cached location information .

A complete list of standard tags available in HTML5 is given below. All the tags are ordered alphabetically along with an indication if they have been introduced newly or they have been deprecated in HTML5.

Tag	Description
<!--...-->	Specifies a comment
<!DOCTYPE>	Specifies the document type
<a>	Specifies an anchor
<abbr>	Specifies an abbreviation
<acronym>	**Deprecated:**Specifies an acronym
<address>	Specifies an address element
<applet>	**Deprecated:** Specifies an applet
<area>	Specifies an area inside an image map
<article>	**New Tag:** Specifies an independent piece of content of a document, such as a blog entry or newspaper article

<aside>	**New Tag:**Specifies a piece of content that is only slightly related to the rest of the page.
<audio>	**New Tag:**Specifies an audio file.
<base>	Specifies a base URL for all the links in a page
<basefont>	**Deprecated:** Specifies a base font
<bdo>	Specifies the direction of text display
<bgsound>	Specifies the background music
<blink>	Specifies a text which blinks
<blockquote>	Specifies a long quotation
<body>	Specifies the body element
 	Inserts a single line break
<button>	Specifies a push button
<canvas>	**New Tag:**This is used for rendering dynamic bitmap graphics on the fly, such as graphs or games.
<caption>	Specifies a table caption
	Deprecated: Specifies centered text
<col>	Specifies attributes for table columns
<colgroup>	Specifies groups of table columns
<command>	**New Tag:**Specifies a command the user can invoke.
<comment>	Puts a comment in the document
<datalist>	**New Tag:**Together with the a new list attribute for input can be used to make comboboxes
<dd>	Specifies a definition description
<del>	Specifies deleted text
<details>	**New Tag:**Specifies additional information or controls which the user can obtain on demand.
<dir>	**Deprecated:** Specifies a directory list
	Specifies a section in a document
<dl>	Specifies a definition list
<dt>	Specifies a definition term
<embed>	**New Tag:**Defines external interactive content or plugin.
<fieldset>	Specifies a fieldset
<figure>	**New Tag:**Specifies a piece of self-contained flow content, typically referenced as a single unit from the main flow of the document.
<b>	Specifies bold text

<big>	**Deprecated:**Specifies big text
<i>	Specifies italic text
<small>	Specifies small text
<tt>	**Deprecated:**Specifies teletype text
<font>	**Deprecated:** Specifies text font, size, and color
<footer>	**New Tag:**Specifies a footer for a section and can contain information about the author, copyright information, et cetera.
<form>	Specifies a form
<frame>	**Deprecated:**Specifies a sub window (a frame)
<frameset>	**Deprecated:**Specifies a set of frames
<head>	Specifies information about the document
<header>	**New Tag:**Specifies a group of introductory or navigational aids.
<hgroup>	**New Tag:**Specifies the header of a section.
<h1> to <h6>	Specifies header 1 to header 6
<hr>	Specifies a horizontal rule
<html>	Specifies an html document
<isindex>	**Deprecated:** Specifies a single-line input field
<iframe>	Specifies an inline sub window (frame)
<ilayer>	Specifies an inline layer
<img>	Specifies an image
<input>	Specifies an input field
<ins>	Specifies inserted text
<keygen>	**New Tag:**Specifies control for key pair generation.
<keygen>	Generate key information in a form
<label>	Specifies a label for a form control
<layer>	Specifies a layer
<legend>	Specifies a title in a fieldset
<li>	Specifies a list item
<link>	Specifies a resource reference
<map>	Specifies an image map
<mark>	**New Tag:**Specifies a run of text in one document marked or highlighted for reference purposes, due to its relevance in another context.
<marquee>	Create a scrolling-text marquee
<menu>	**Deprecated:** Specifies a menu list

<meta>	Specifies meta information
<meter>	**New Tag:**Specifies a measurement, such as disk usage.
<multicol>	Specifies a multicolumn text flow
<nav>	**New Tag:**Specifies a section of the document intended for navigation.
<nobr>	No breaks allowed in the enclosed text
<noembed>	Specifies content to be presented by browsers that do not support the <embed>tag
<noframes>	**Deprecated:**Specifies a noframe section
<noscript>	Specifies a noscript section
<object>	Specifies an embedded object
<ol>	Specifies an ordered list
<optgroup>	Specifies an option group
<option>	Specifies an option in a drop-down list
<output>	**New Tag:**Specifies some type of output, such as from a calculation done through scripting.
<p>	Specifies a paragraph
<param>	Specifies a parameter for an object
<cite>	Specifies a citation
<code>	Specifies computer code text
<dfn>	Specifies a definition term
	Specifies emphasized text
<kbd>	Specifies keyboard text
<samp>	Specifies sample computer code
	Specifies strong text
<var>	Specifies a variable
<plaintext>	**Deprecated:** Render the raminder of the document as preformatted plain text
<pre>	Specifies preformatted text
<progress>	**New Tag:**Specifies a completion of a task, such as downloading or when performing a series of expensive operations.
<q>	Specifies a short quotation
<ruby>	**New Tag:**Together with <rt> and <rp> allow for marking up ruby annotations.
<script>	Specifies a script
<section>	**New Tag:**Represents a generic document or application section.

<select>	Specifies a selectable list
<spacer>	Specifies a white space
	Specifies a section in a document
<s>	**Deprecated:** Specifies strikethrough text
<strike>	**Deprecated:** Specifies strikethrough text
<style>	Specifies a style definition
<sub>	Specifies subscripted text
<sup>	Specifies superscripted text
<table>	Specifies a table
<tbody>	Specifies a table body
<td>	Specifies a table cell
<textarea>	Specifies a text area
<tfoot>	Specifies a table footer
<th>	Specifies a table header
<thead>	Specifies a table header
<time>	**New Tag:**Specifies a date and/or time.
<title>	Specifies the document title
<tr>	Specifies a table row
<u>	**Deprecated:** Specifies underlined text
<ul>	Specifies an unordered list
<video>	**New Tag:**Specifies a video file.
<wbr>	**New Tag:**Specifies a line break opportunity.
<wbr>	Indicate a potential word break point within a <nobr> section
<xmp>	**Deprecated:** Specifies preformatted text